LS-CAT Management Board Meeting

Keith Brister
May 16, 2013

LS-CAT Management Board Meeting Agenda

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       Start Time: 1 PM CDT (2 PM EDT) May 16, 2013
            Audio: 1-888-850-4523
         Passcode: 190580
          Video 1: http://conf1.ls-cat.org
          Video 2: http://conf2.ls-cat.org
Physical Location: 9700 S. Cass Ave Bldg 436A, Argonne, IL
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- Approval of last meeting's minutes
- Status
  - New MD-2 control code (implementing mid to late June)
  - New Robot software (F installed, other stations later)
- Genentech
- Data Collection/Analysis Workshop 7/26-7/27 @ LS-CAT
- Repolishing of 21-ID-D Vertical Mirror
  - Request vote to spend about $100K
- Budget
  - Flat, +3%, or + 25K/share?
  - Estimated capital fund of $1.9M at start of FY14
- New Detector Grant: due 9/13/13
- New LS-CAT Scientific Director (Wayne is stepping down)
  - Not required to be from NU but has been advantageous
    - LS-CAT operated by NU SRC (new co-director TBA)
  - Cannot be paid if S.D. is a member of the board
  - Meetings: 4 PUC + 4 LSC + 2 MB + 1 SRC per year
- APS Review of Sector 21
- life sciences upgrade projects in the context of the APS-U project
- New Business

LS-CAT Status in a Nutshell

- New code to increase data collection speed by 10 to 20 percent is being implemented this down period and will be made available to users starting mid to late June. Code has been tested (and is being used) by NE-CAT in an interesting cross-CAT collaboration.

- New robot software has been installed at 21-ID-F and will be propagated to the other stations when it is logistically expedient to do so. (Previous attempt to upgrade all stations last Fall was disastrous) Users should see an improved sample loading time of 10 to 20 percent.

Genentech

Genentech has requested Associate CAT Member status to gain access to beamtime similar to Emerald's access. That is, to a pool of beamtime made available by those members who have chosen to participate. They plan to start off with a commitment of 60 hours per year (small by any standards) but will likely increase this as time goes on.

A vote now would set things in motion.

LS-CAT Data Collection and Structure Solution Workshop

LS-CAT would like to invite the members to a data collection/ structure solution/ refinement/ building hands-on workshop starting July 26 and July 27. The workshop will start with hands-on data collection and processing at the LS-CAT beamlines, participants must bring their most challenging projects to work with the staff. The following day, the workshop participants will listen to lectures/tutorials from the staff at LS-CAT.

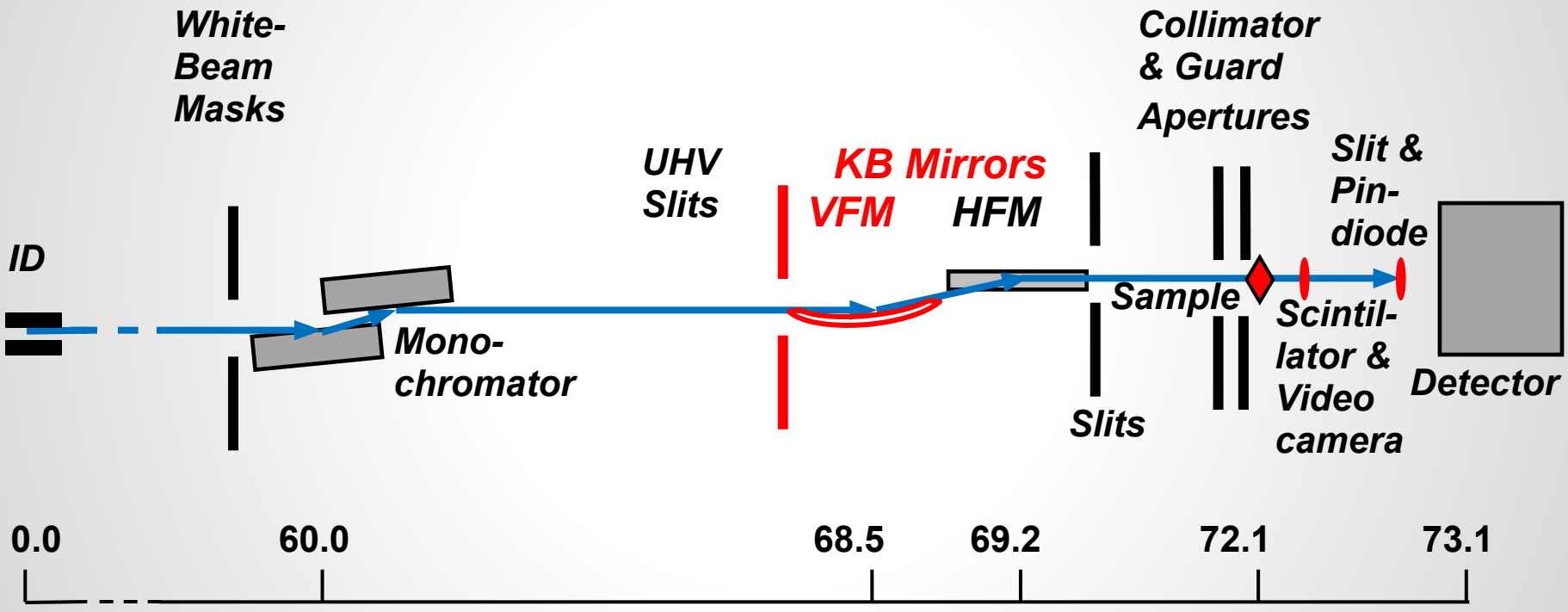
7/26 Friday -- 9:00 - 22:00 Data Collection & Data Processing @ LS-CAT 21 D, F, & G

- Data Processing and Analysis
  - Data Collection Theory and Strategy
  - Data Processing with HKL
  - Data Processing with Xia2/XDS/Aimless
  - Data Processing with AutoProc/XDS/SCALA
  - Data Analysis with Xtriage
- Data Collection -- Remote Access and/or Live
  - Remote Access
  - Real time data collection at stations D, F, & G
  - Data processing at the beamline

7/27 Saturday 9:00-21:00

- Phasing
  - Anomalous and/or Isomorphous Replacement Phasing
  - Molecular Replacement Phasing
- Refinement
  - Refmac
  - Phenix.refine
- Auto and Manual Building
  - ARP/wARP
  - Phenix.autobuild
  - Coot
  - Structure Validation

LS-CAT 21ID-D beamline layout



Bimorph mirrors

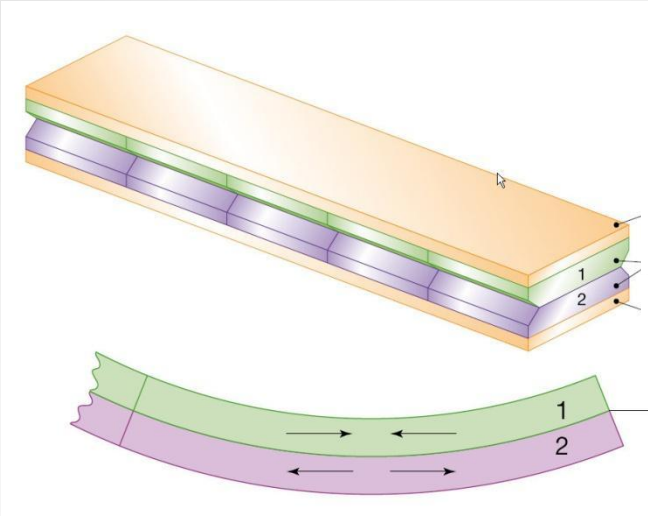


Fig 1. From FMB –Oxford Mirror Systems Catalogue 2013
http://www.fmb-oxford.com

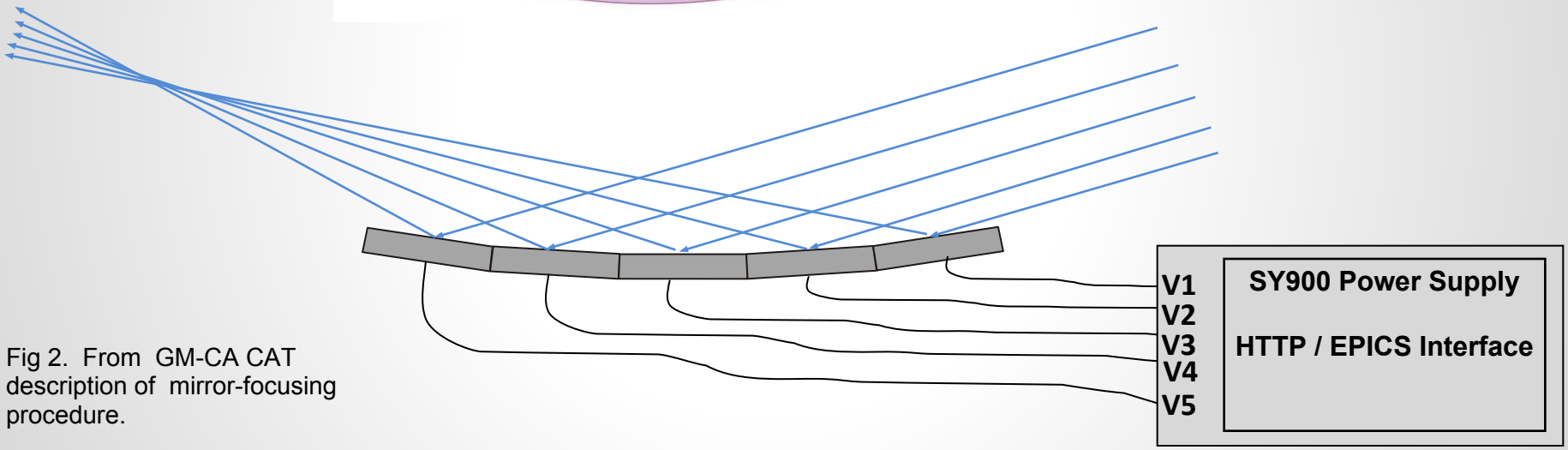


Fig 2. From GM-CA CAT description of mirror-focusing procedure.

Advantages: **multiple focus adjusting parameters *vs* 1 or 2 with a bender**

Disadvantages: **very slow, cross-talk between segments, design flaws**

Bimorph Mirror



LS-CAT Problems With D-beamline Vertically-Focusing Mirror (VFM)

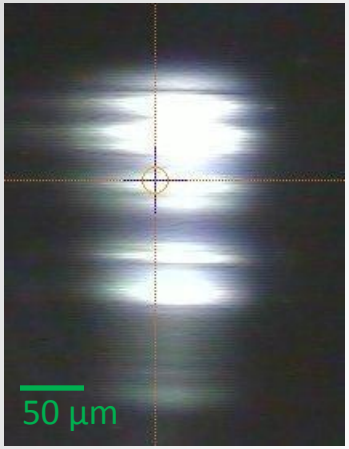


Fig. 1 X-ray beam image at sample position:
0 V on electrodes,
En=8 keV,
Lane=glass,
α = 3.5 mrad,
all slits widely open.

Problem 1.

Reflected beam is not naturally focused at the sample position as required by the mirror design. (Fig.1).

Possible reasons:

(a) Error in the prefiguring radius of the mirror R.

R=2q/sinα

q - required focal distance, *α* - beam incidence angle,

p - mirror position from the source

Design: p = 68.5 m, **q = 3.5 m**, α = 3.5 mrad, **R = 2.0 km**

APS measurements: **R = 1.65** ±0.4 km -> **q = 2.89** m

while **q=3.6 m** at the D-beamline

(b) Mirror-surface figure is affected by clamping to support and counterweights attached to the mirror frame.

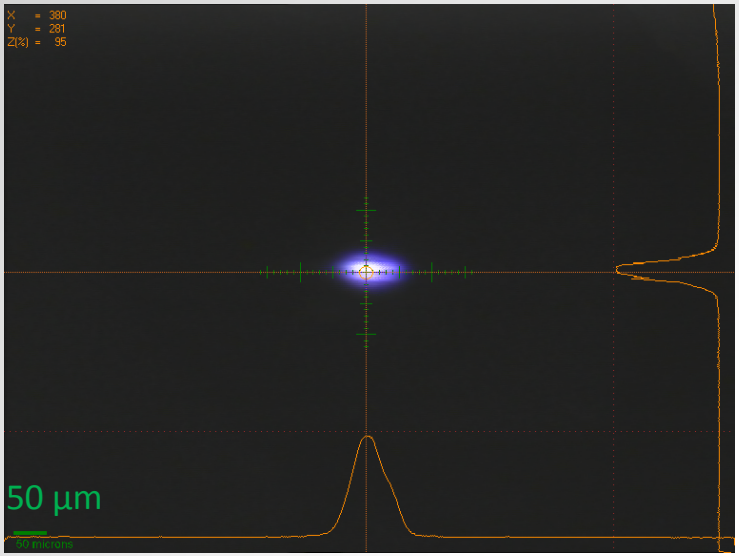


Fig. 2a

Problem 2.

We can hardly get 20 µm beam, but rather 30 µm at the sample position (Fig. 2a)

Problem 3.

Beam focused to 30 µm at 700 mm downstream of the sample (Fig. 2b) does not provide a smooth 70-100 µm profile at the sample position (Fig. 2c) that is required by the everyday operation on the D-beamline.

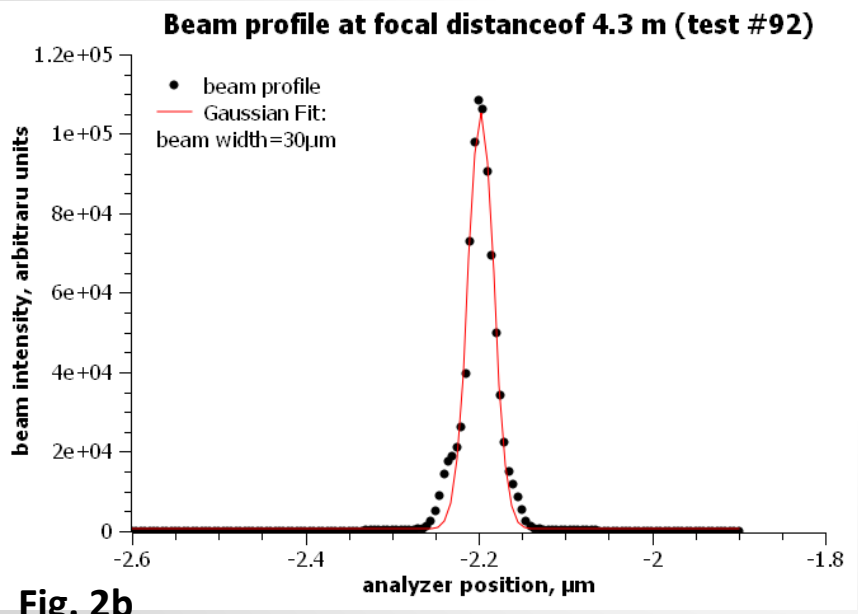


Fig. 2b

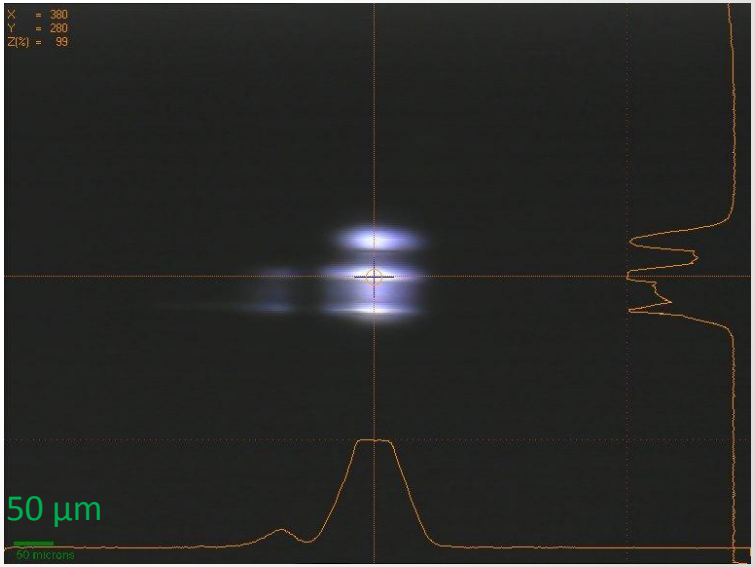
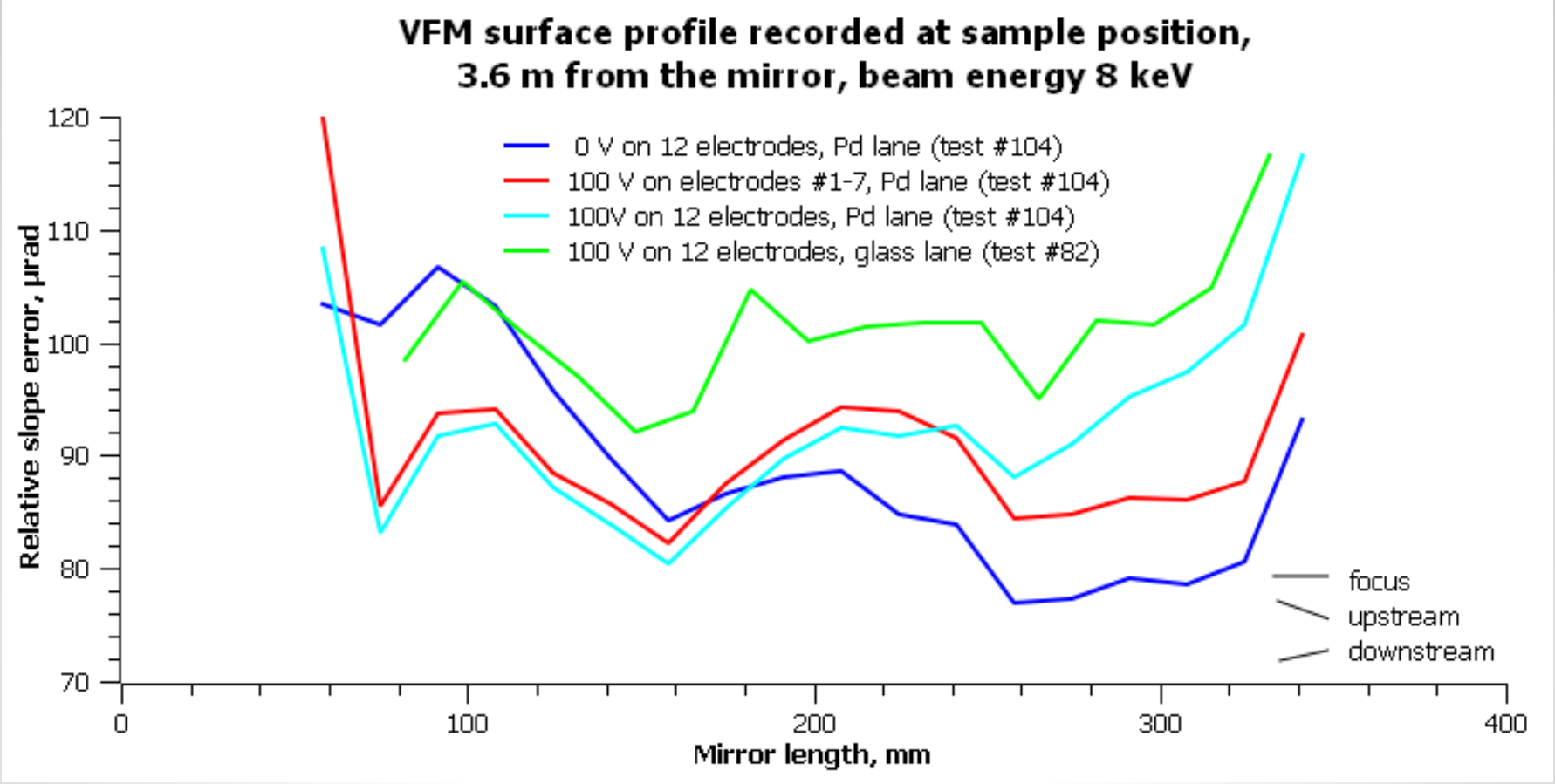


Fig. 2c



Problem 4.

Slope error of the middle part of the mirror surface was found to be ~10 times higher than 1.5 µrad that is in the mirror specs. It is even higher for the mirror end segments (not shown on the graph).

Problem 5.

Focusing of the beam deteriorates at changing lanes on the mirror which is required at changing beam energy.

This can significantly slow down the beamline operation.

X-ray mirrors refurbishment

BIMORPH MIRRORS: THE GOOD, THE BAD, AND THE UGLY

Nuclear Instruments and Methods in Physics Research. A 710 (2013) 87

Bimorph mirrors are employed on numerous synchrotron X-ray beamlines. **However, many such beamlines were not achieving the desired size and shape of the reflected X-ray beam**.

The biggest problem is the **"junction effect"**: corrugations on the optical surface which coincide with the location of junctions between piezo ceramics. Leading candidate for the cause of the damage is the **glue which bonds the piezo ceramics** together and to the optical substrate and cures over several weeks changing on exposure to ultra-high vacuum or synchrotron radiation.

Periodic variation in radius of mirror curvature was also attributed to the **solid electrical connectors** that can act as springs and oppose changes to the bending.

Three pairs of bimorph mirrors, operational since 2007 on the PX beamlines at Diamond synchrotron, UK, were re-polished at Thales-SESO. All evidence of unwanted structure was removed and has not re-appeared within one year of beamline operations.

Our proposal: To send VFM-E to SESO for refurbishment in order to later replace VFM-D with it.

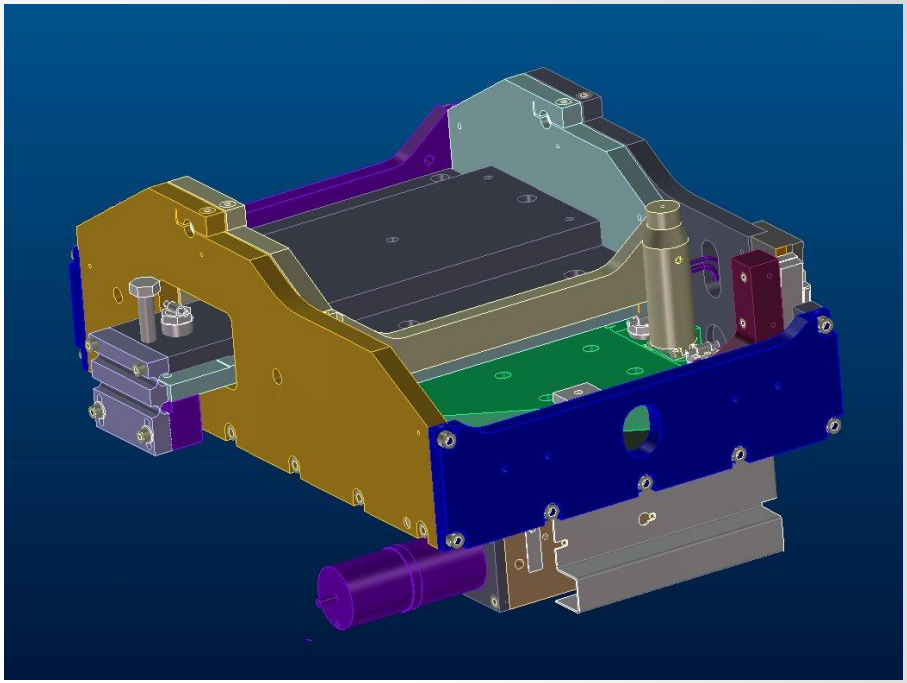
New VFM Mechanics

Upgrades the vertical focusing mirror in 21-ID-D to the latest design for improved stability.

Current design shows a strong resonance at about 60 Hz causing the beam to bounce up and down unacceptably. Our fix of adding mass to the mirror assembly to move the resonance helps enough for the mirror to become useful but it is still not optimal.

Replacing the mechanics should solve this instability.

As with the re-polishing, we can send back the 21-ID-E mechanics to be upgraded to the new design and just swap them out with the 21-ID-D mechanics during a down period.



D Mirror Vote

There are two jobs we'd like to perform on the D mirror system. These can be done using the E mirror and E mirror mechanics and then simply exchange them with the D system when ready (probably in the January 2014 down period). Because we can perform the work on the E mirror system there will be no down time.

Repolishing the D vertical focusing mirror will run approximately $70K. This will address the issues identified by Elena.

Replacing the D vertical focusing mirror mechanics will likely run approximately $30K. This should address the vibrational instability.

A vote is requested specifically authorizing me to proceed with the D Vertical Focusing Mirror work. This money has already be included in the FY13 estimated spending shown on the next slide.

LS-CAT FY14 Proposed Budget(s)

Description	FY2013 Total (Est)	FY2013 Budget	FY 14 FLAT	FLAT-FY13	FY 14 3%	3% - FLAT	FY 14 325	325 - FLAT
Salaries	858,977	847,424	878,133	30,709	878,133		878,133	
Fringe Benefits	233,600	228,488	251,146	22,658	251,146		251,146	
Supplies	74,291	76,000	76,000		76,000		85,000	9,000
Supplies, Restricted	1,409	5,000	2,000	-3,000	2,000		2,000	
Services	12,782	66,000	13,000	-53,000	13,000		13,000	
Services, Restricted	15,813	12,000	17,000	5,000	17,000		17,000	
External Space Rental	109,935	109,935	113,233	3,298	113,233		113,233	
Travel, Domestic	7,321	2,000	5,000	3,000	5,000		20,000	15,000
Utilities & Telephone	9,473	8,000	8,000		8,000		8,000	
Capital Equipment	269,108	208,000	197,655	-10,345	256,155	58,500	328,643	130,988
Indirect Cost - Sponsored	319,101	387,153	388,833	1,680	388,833		396,345	7,512
Total	1,911,809	1,950,000	1,950,000		2,008,500	58,500	2,112,500	162,500
Per Share	294,124	300,000	300,000		309,000		325,000	

Estimated capital fund at start of FY14: $1.9M. The LS-CAT fiscal year starts September 1.

Assuming a constant repair rate, a flat budget will require that we start to deplete the capital fund in FY14. If we get lucky (or foolishly stop repairing things) we can perhaps delay this to FY15.

The current LS-CAT inflation rate is about 3% (nationally it was 1.5% in March) and a 3% budget increase will make it less likely we'd need to spend more than our annual budget. A further increase should be expected every year from now on if we only see a increase 3% for FY14.

With a budget of $325K/share we should be able to keep the dues constant until the start of the next LS-CAT agreement (March 1, 2017).

Constant $325K/Share through 2017

Salaries			100% KB, JA, MB, JVO, DS, SA 90% JSB, NB
	Base	878,133	
	Fringe	251,146	
	Total		1,129,279
Items that attract no F&A costs			
	Capital Equip.	328,643	
	Rent	113,233	
	Telcom	8,000	
	Total		449,876
Non-Salary Items the attract F&A costs			
	Lab Supplies	85,000	
	Office Supplies	2,000	
	Lab Services	13,000	
	Insurance	17,000	
	Travel	20,000	
	Total		137,000
	F&A		396,345
Total			2,112,500
Per Share			325,000
Per 0.5 Share			162,500

Note: Salaries are only in the planning stage at this point. Any increase or decrease between the planned salaries and the actual salaries will be made up using the capital equipment line and will not change the amount billed.

Bills are due and payable September 1, 2013. Payment arrangements with NU must be made by September 30, 2013 to avoid being declared delinquent.

Vote 1: Capitalization of maintenance when possible:

Pre-approval is requested to use the capital equipment line for purchase of replacement equipment required to maintain the facility.

Vote 2: The amounts. The "per share" amount applies to NU, MSU, UM, VARI, and WSU while the "Per 0.5 Share" amount applies to UI, VU, and UW. Amounts received by NU for the pooled ACM program in FY13 will be applied to FY14 dues to contributing members (currently WSU).

Inflation Adjusted Annually (3% FY14)

Salaries			
	Base	878,133	
	Fringe	251,146	
	Total		1,129,279
Items that attract no F&A costs			
	Capital Equip.	256,155	
	Rent	113,233	
	Telcom	8,000	
	Total		377,388
Non-Salary Items the attract F&A costs			
	Lab Supplies	76,000	
	Office Supplies	2,000	
	Lab Services	13,000	
	Insurance	17,000	
	Travel	5,000	
	Total		113,000
F&A			388,833
Total			2,008,500
Per Share			309,000
Per 0.5 Share			154,500

100% KB, JA, MB, JVO, DS, SA
90% JSB, NB

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Flat Budget: Dip into capital fund

Salaries			100% KB, JA, MB, JVO, DS, SA 90% JSB, NB
	Base	878,133	
	Fringe	251,146	
	Total		1,129,279
Items that attract no F&A costs			
	Capital Equip.	197,655	
	Rent	113,233	
	Telcom	8,000	
	Total		318,888
Non-Salary Items the attract F&A costs			
	Lab Supplies	76,000	
	Office Supplies	2,000	
	Lab Services	13,000	
	Insurance	17,000	
	Travel	5,000	
	Total		113,000
	F&A		388,833
Total			1,950,000
Per Share			300,000
Per 0.5 Share			150,000

Note: Salaries are only in the planning stage at this point. Any increase or decrease between the planned salaries and the actual salaries will be made up using the capital equipment line and will not change the amount billed.

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Pre-approval is requested to use the capital equipment line for purchase of replacement equipment required to maintain the facility.

Vote 2: The amounts. The "per share" amount applies to NU, MSU, UM, VARI, and WSU while the "Per 0.5 Share" amount applies to UI, VU, and UW. Amounts received by NU for the pooled ACM program in FY13 will be applied to FY14 dues to contributing members (currently WSU). **This budget will likely require spending saved capital equipment monies to support operations and approval for these amounts also approves spending more than we are billing.**

New detector grant

Although the previous detector grant application (Alfonso Mondragon, PI) was unsuccessful we should try again. The deadline for the current round of High-end Instrumentation Grants is September 13, 2013. Likely we'll want to update and rewrite the proposal making sure that all issues are addressed.

Over all at the APS it seems that fewer than half of such grants are being funded and that multiple attempts are required. There does not appear to be a rhyme or reason to the selection process.

We should consider using some of our $1.9M of saved capital to fund a new detector regardless of the NIH detector grant status. (A fast readout Rayonix detector runs about $1M but such a detector has yet to be delivered.)

6.0 Scientific Director of the LS-CAT

6.1 The initial Scientific Director of the CAT will be a scientist of NU who will guide construction during Phase I and acquisition of funding for construction, which is anticipated to last approximately 3 years, and scientific development of subsequent phases. Thereafter, the Scientific Director of the CAT will be appointed by the Board and will be responsible to the Board for tasks including, but not limited to, those listed in paragraph 6.2. The Director shall coordinate all functions with the Board through its Chair.

6.2 The Director will be empowered to:

6.2.1 Serve as the public spokesperson for the CAT;

6.2.2 Serve with the Manager as the liaison between the CAT and the APS;

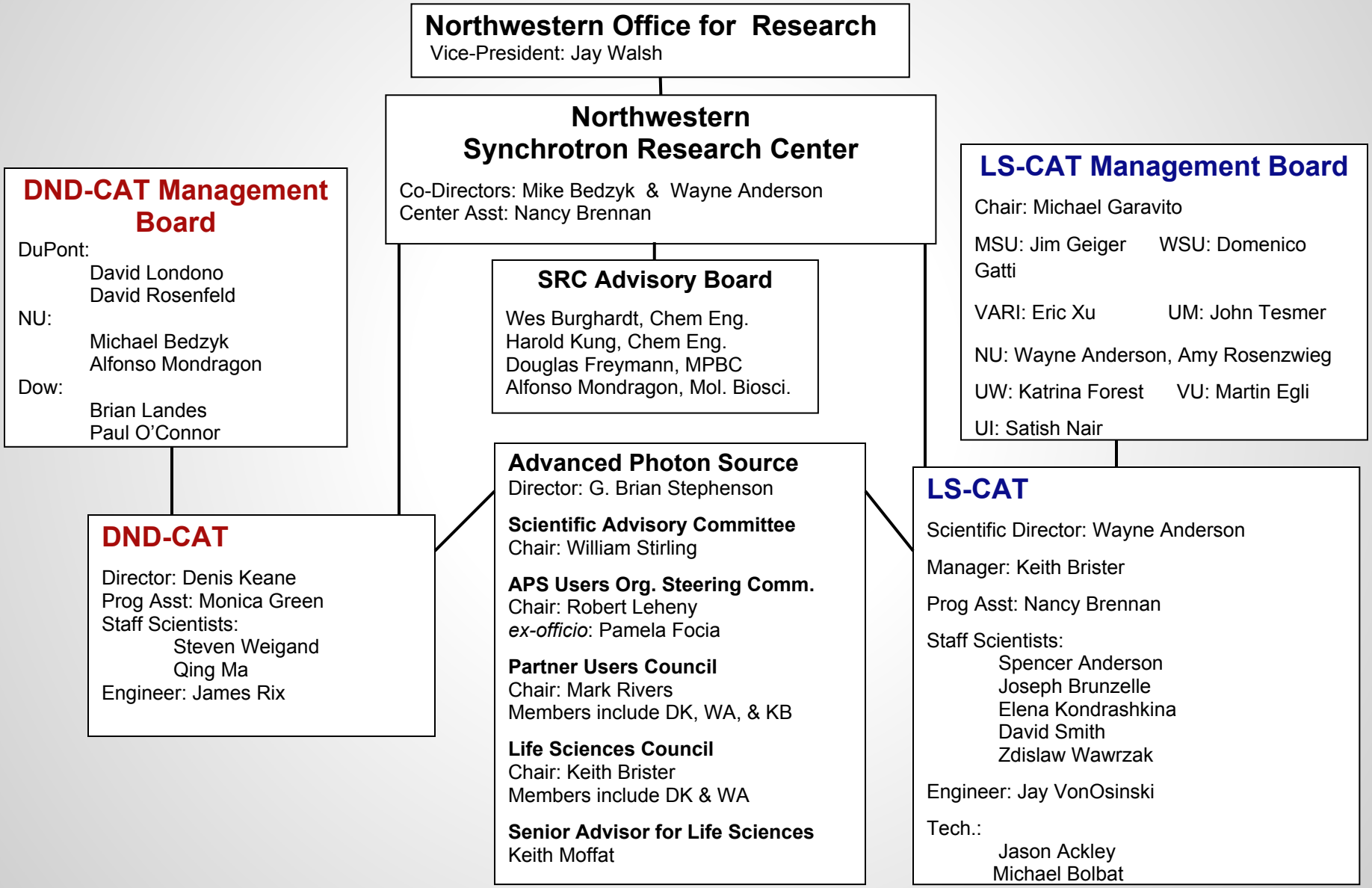
6.2.3 Coordinate preparation and submission of proposals or contracts with the Board to provide funding for the development and upgrading/maintenance of CAT facilities beyond Phase I;

6.2.4 Fulfill any required reporting functions of the CAT; and

6.2.5 Perform such other duties as directed by the Board.

6.3 The Director will not be liable for any actions taken within the scope of the Director's official CAT authority and will be indemnified by all Parties equally against the same. Excluded from indemnity are acts involving intentional misconduct, acts outside the scope of the Director's authority, willful violations of law, and transactions from which the Director may derive an improper personal benefit.

The (Simplified) LS-CAT/DND-CAT Org Chart



APS Review of LS-CAT

Originally scheduled for this Fall it looks now like it will now take place in November 2014. In preparation for this we'll start now. We will contact you for the following:

1. Complete LS-CAT publication list
2. List of theses
3. 2 page "Research Highlights" from your lab
4. Future directions for your lab
5. Future needs from LS-CAT

We'll ask some of the PI's to come and give a presentation to the SAC review committee

The SAC Recommendations

March 2011

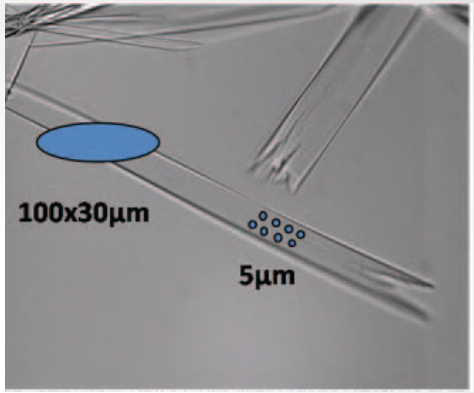
A1. Very Strongly Recommended

CDR	Title	Panel	Priority
4.2.2	SPX Hard X-ray - Diffraction & Imaging	1	A1
4.2.2	SPX Hard X-ray - Spectroscopy	1	A1
4.3.2	Wide-Field Imaging Beamline	4	A1
4.3.4	High-Energy Tomography	4	A1
4.3.7,4.3.8	In Situ Nanoprobe/Cryonanoprobe (NGN)	4	A1
4.4.4	Resonant Inelastic X-ray Scattering (MERIX)	3	A1
4.5.4	High-Energy Diffraction	2	A1
4.5.5	Magnetic Spectroscopy	3	A1
4.6.2	XIS - Tunable ID Beamlines	2	A1
4.6.4	Micro and 3D Diffraction	2	A1
4.7.3	Cryo Sample Preparation Facility	5	A1
4.7.4	Enhanced SAXS/WAXS	5	A1
4.7.5	Microfocus MX Beamline	5	A1
4.7.6	Enhanced Pump/Probe for Physical Sciences	1	A1
4.7.6	Enhanced Time-Resolved MX Beamline	5	A1

A2. Strongly Recommended

CDR	Title	Panel	Priority
4.2.2	SPX-ray Facility Soft X-ray Beamline	1	A2
4.3.3,4.3.7	TXM	4	A2
4.3.5	XPCS and Coherent GIXS	2	A2
4.3.6	Fluid Dynamics Imaging Beamline	4	A2
4.4.2	Advanced Spectroscopy Beamline	1	A2
4.4.3	LERIX-2 Beamline	1	A2
4.5.2	High-Magnetic-Field Scattering	3	A2
4.5.3	High Pressure Studies Sub-micron Beams	3	A2
4.6.2	XIS - Fixed Angle ID Beamlines	2	A2
4.6.5	Resonant Interface Scattering	2	A2
4.7.2	BioNanoprobe	5	A2
4.X.X	Long Wavelength PX Beamline	5	A2

Images from the Connecting Proteins to Organisms section of the APS CD-0 Report, May 2009

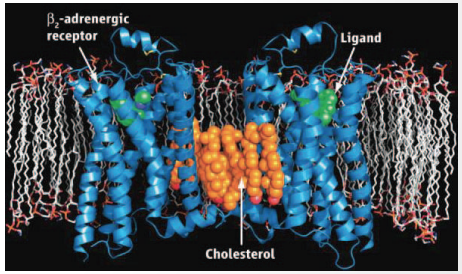


Courtesy B. Koblika and W. Weis
Stanford University

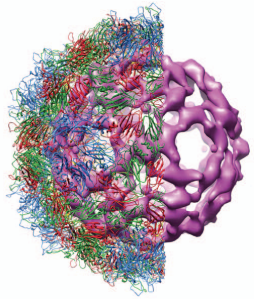


THE 2012 NOBEL PRIZE IN CHEMISTRY

Robert J. Lefkowitz, Brian K. Kobilka



Cherezov, V., High-resolution crystal structure of an engineered human beta2-adrenergic G protein coupled receptor, Science, **318,** 1211 (2007).



Venkataram, S., Reddy, S.P., Loo, J., et al., Structure of Seneca Valley Virus-001: An Oncolytic Picornvirus Representing a New Genus, Structure, **16**, 1555 (2008).

The APS Life Sciences Council

Subcommittee of the Partner User Council

Represents the needs of the LS community as a whole

12 Sectors

>50 Institutions

Over 50% of APS experiments in 2012 were done on LSC participating beamlines

SBC-CAT Sector 19
GMCA @ APS Sector 23
Sector 2 (XRM)
Sector 12 (Bio SAXS/WAXS)

Diversity of Funding of MX CATs

BioCARS
NIH (NIGMS)
(APS for non-biological TR studies)

SBC
DOE, BER

GM/CA-CAT
NIH (NIGMS, NCI)

LS-CAT
University of Michigan
Michigan State University
Van Andel Institute
Wayne State University
Vanderbilt University
University of Illinois
University of Wisconsin/Madison
Northwestern University
GlaxcoSmithKline
Emerald Biostructures
Cayman Chemical
Grand Valley State University
Rice University
Medical College of Wisconsin

IMCA-CAT
Abbot
Bristol-Myers Squibb
Johnson & Johnson (Subscriber)
Merck Research Laboratories
Novartis
Pfizer
HWI

LRL-CAT
Eli Lilly & Company

Aggregate MX CAT Investment

- Cap. Eq. since 2010 $ 11M
- Total capital equipment $ 132M
- Annual operating costs $ 19M

DOE-BES does not fund MX at APS

NE-CAT
NIH (NIGMS)
Columbia University
Cornell University
Harvard University
Memorial Sloan-Kettering Cancer Center
Massachusetts Institute of Technology
Rockefeller University
Yale University

SER-CAT
Duke University
Emory University
Florida State University
Georgia State University
Georgia Tech Research Corporation
Medical University of South Carolina
Monsanto Company
NIH Intramural Research Program
North Carolina State University
Rosalind Franklin U. of Med. and Sci.
Scripps Florida
St. Jude Children's Research Hospital
University of Alabama at Birmingham
University of South Florida
University of Georgia
University of Kentucky
University of Missouri at Kansas City
University of North Carolina at Chapel Hill
University of Pittsburgh
University of South Carolina
University of Virginia
Vanderbilt University

Diversity of Funding of MX CATs

BioCARS
NIH (NIGMS)
(APS for non-biological TR studies)

SBC
DOE, BER

GM/CA-CAT
NIH (NIGMS, NCI)

LS-CAT
University of Michigan
Michigan State University
Van Andel Institute
Wayne State University
Vanderbilt University
University of Illinois
University of Wisconsin/Madison
Northwestern University
GlaxcoSmithKline
Emerald Biostructures
Cayman Chemical
Grand Valley State University
Rice University
Medical College of Wisconsin

IMCA-CAT
Abbot
Bristol-Myers Squibb
Johnson & Johnson (Subscriber)
Merck Research Laboratories
Novartis
Pfizer
HWI

LRL-CAT
Eli Lilly & Company

Aggregate MX CAT Investment

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DOE-BES does not fund MX at APS

NE-CAT
NIH (NIGMS)
Columbia University
Cornell University
Harvard University
Memorial Sloan-Kettering Cancer Center
Massachusetts Institute of Technology
Rockefeller University
Yale University

SER-CAT
Duke University
Emory University
Florida State University
Georgia State University
Georgia Tech Research Corporation
Medical University of South Carolina
Monsanto Company
NIH Intramural Research Program
North Carolina State University
Rosalind Franklin U. of Med. and Sci.
Scripps Florida
St. Jude Children's Research Hospital
University of Alabama at Birmingham
University of South Florida
University of Georgia
University of Kentucky
University of Missouri at Kansas City
University of North Carolina at Chapel Hill
University of Pittsburgh
University of South Carolina
University of Virginia
Vanderbilt University

IL MI TN WI WA TX NY MA NC GA FL SC AL KY MO PA VA
~34 Senators
~44 Representatives

Cryo Prep Facility (Chris Jacobsen)

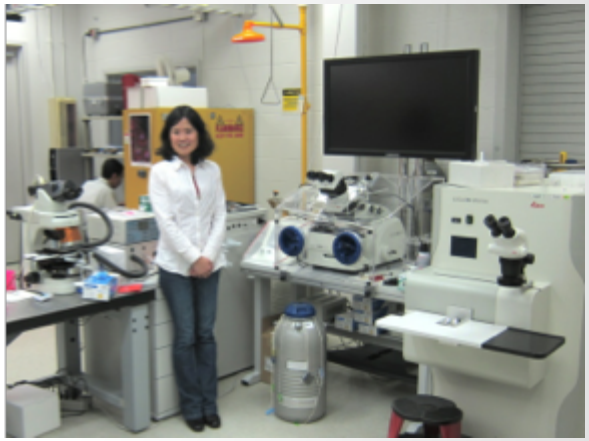
A support facility for biological samples LS-CAT bionanoprobe at Sector 21, the microprobe at Sector 2, and likely other programs working with cryo samples.

Equipment:

$4.2M +/- $0.5M

Operations:

$1.0M +/- $0.2M Annual

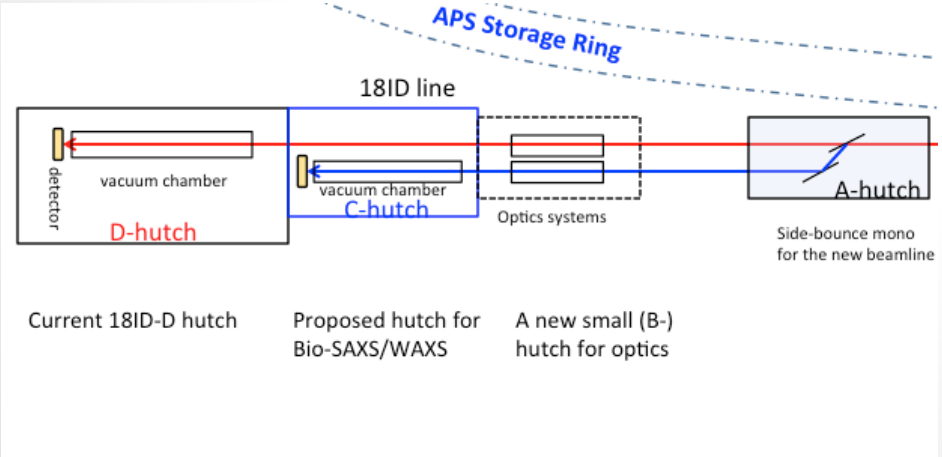


Bio SAX/WAX (Yun-Xing Wang)

Possibly sited at Sector 18. The current concept is to cant the existing sector and build new hutches to accommodate the new SAX/WAX capabilities. Using a layout similar to Sector 12 it should be possible to install a large offset monochromator to gain enough distance to place a new experimental hutch for this new facility.

This project currently needs help with the layout and costing. It's not clear who would operate and maintain the resulting beamline.

Let's take a stab in the dark and say this will end costing $8M +/- $2M. That's probably not too far off the mark. Additionally, it will probably cost something like $1.0 M +/- 0.2 M annually to operate.



Microfocus MX beamline (Andrzej Joachimiak)

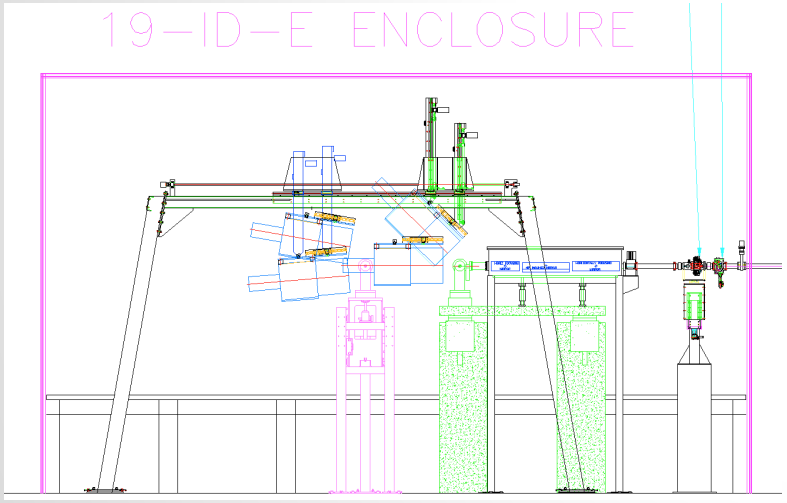
Plans have been developed with layouts and preliminary costing. Sector 19 (SBC-CAT) would be canted and a bounce up large offset mono would raise the beam above the existing endstation equipment. This beamline is expected run about $9M +/- $1M.

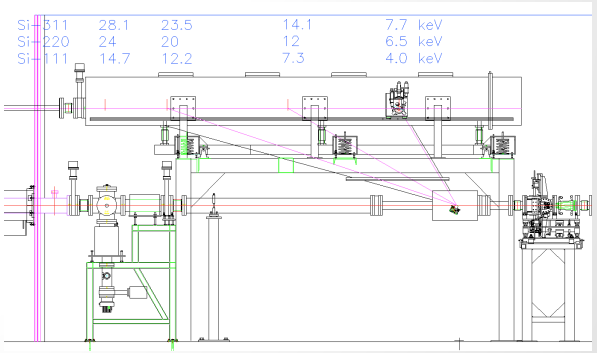
At Sample:

$2.2X10^{11}$ photons/sec

$9.8X10^{16}$ photons/sec/mm2

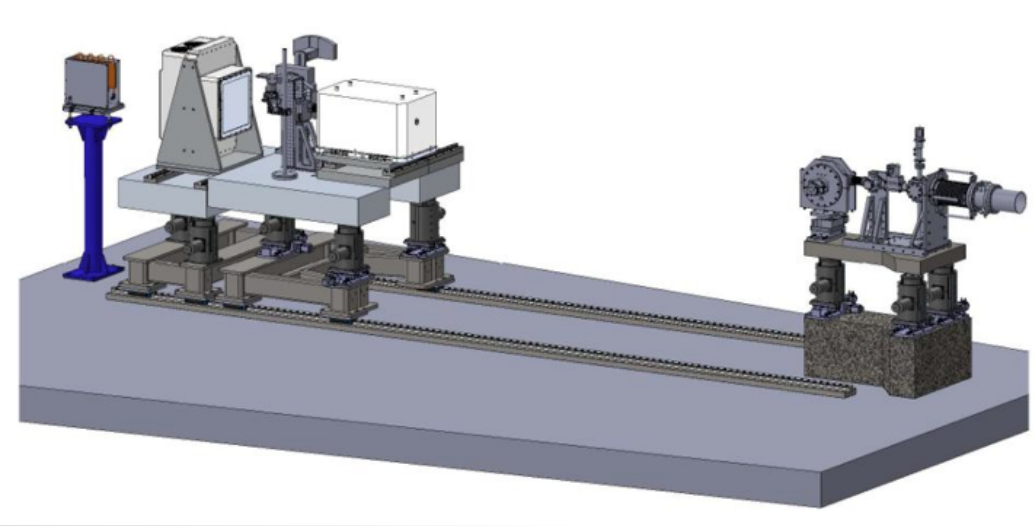
2.4µm by 0.7µm FWHM





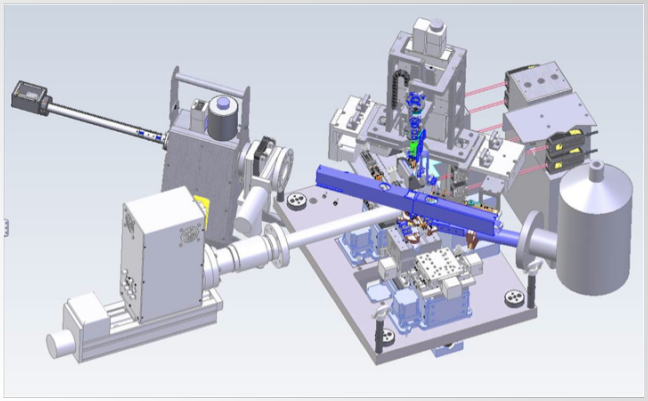
Enhanced time resolved MX beamline (Keith Moffat)

This is related to the "Enhanced Pump/Probe for Physical Sciences" project that has been proposed to be part of the baseline scope of the APS-U. Some funding has been requested from the NIH to support the MX portion.



- $1x10^{10}$ photons/pulse in 24-bunch mode (79 ps resolution)
- $3.8x10^{10}$ photons/pulse in hybrid mode (118 ps resolution)

Bionanoprobe (Gayle Woloschak)



There are a several configurations one might consider:

1) BNP II: New tandem undulator beamline. ($12M +/- $2M)

2) BNP II Alternative: Third canted beamline on Sector 21 with a dedicated bionanoprobe station ($9M +/- $4M: Savings in shielding would possibly be offset by engineering costs for this complicated design.)

3) BNP++: Third undulator in a "long straight section" Sector 21 where two undulators feed 21-ID-D (where the bionanoprobe currently runs 25 to 50 percent of the time). Proper selection of undulators would extend the energy range of the instrument and give more flux around 10KeV where the probe is mostly used. ($2M +/- $1M)

4) BNP++ Alternative: Use a revolver to provide most of the benefits of a third undulator without needed to go to a long straight section configuration. ($2.5M +/- $0.5M). Note that the bionanoprobe would benefit greatly from a decrease in the horizontal emittance.

We'll use configuration 4 to proceed with our estimates although the proposers would really like to see configuration 1.

Extended wavelength beamline (B.C. Wang)

This beamline has a very well developed scientific case including a dedicated workshop during the 2011 APS Users Meeting and a session at the 2012 ACA meeting in Boston. The next step is to propose potential beamline configurations.

For our purposes now we can make some guesses as to what the costs will look like:

One end of the cost spectrum would be an optimization of the beam delivery system and end station on 22ID to provide enhanced capabilities to support a pilot program to develop both the capability and user community, say $3M +/- $1M.

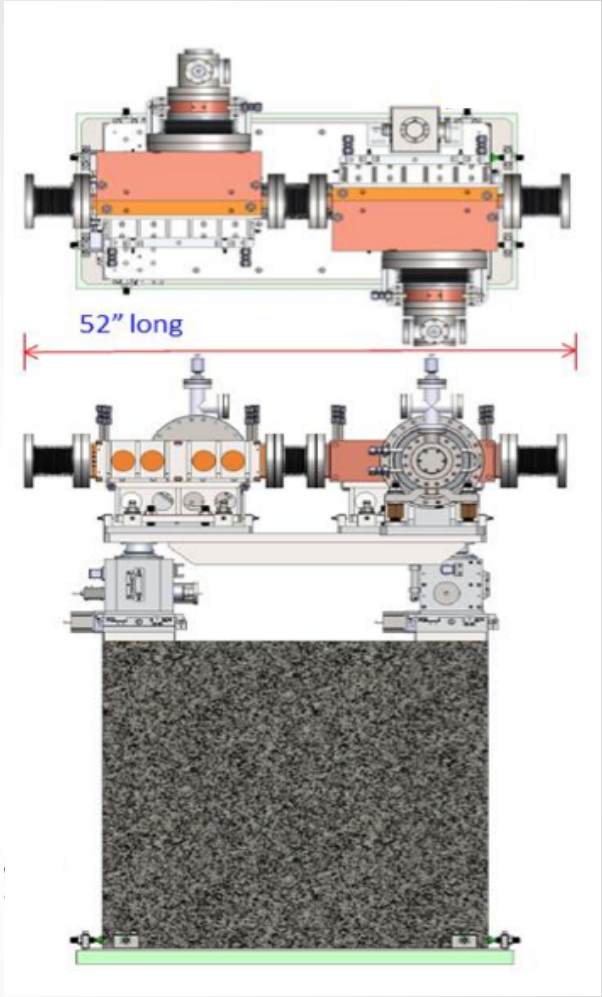
On the other end would be a canted undulator (on Sector 22), a large offset mono, new station(s), and new end station equipment. Say $15M +/- $3M since the specialized detector will not be cheap.

So lets go with $11M +/- $4M as a starting point for planning purposes.

GRID BPMs

At $0.5M per canted sector these BPM's are out of reach for most life sciences beamlines. Although they promise to provide extreme low frequency stability (making facility managers happy) they will not improve data quality and this fact makes them almost impossible to fund at the CAT level.

Sector	Cost ($M)	Uncertainty ($M)
17	0.3	0.1
18	0.3	0.1
19	0.3	0.1
21	0.5	0.1
22	0.3	0.1
23	0.5	0.1
24	0.5	0.1
31	0.3	0.1
Total	**3.0**	**0.8**



Area Detectors

Upgrading the detectors on the protein crystallography beamlines is a simple, straightforward, way to leverage the APS-U advances. Moving most of the crystallography beamlines to pixel array detectors (Dectris or ADSC) or to the latest generation of fast CCD detectors (Rayonix), is required if the APS is to maintain its lead in this field.

The table below estimates the needs to upgrade existing beamlines that have not yet been upgraded.

The cost uncertainty is partially due to unknown future prices and partially due to different detector needs (not all beamlines need the same configuration), hence, the net uncertainty is neither as pessimistic a straight sum nor as optimistic as a quadrature sum.

Sector	# New Detectors	Approx Cost ($M)	Approx Cost Uncertainty ($M)
14	1	2.0	0.5
18	1	2.0	0.5
19	2	4.0	1.0
21	2	4.0	1.0
22	1	2.0	0.5
23	1	2.0	0.5
24	1	2.0	0.5
31	1	2.0	0.5
Total	10	20.0	3.0





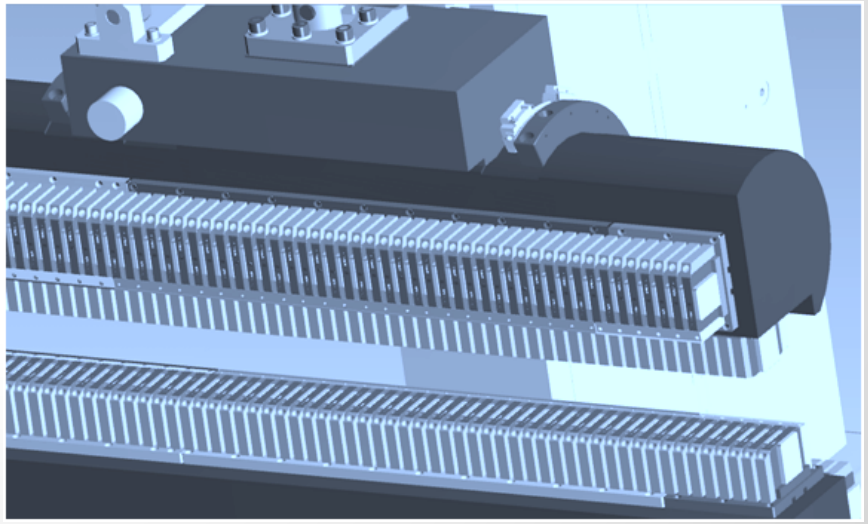
Optimized Undulators

A recent survey of the life sciences beamlines showed 2 beamlines that believe a revolver style undulator would be a better match for their science and 3 other beamlines that believed their programs would be improved with a different undulator from what they have now. Of these, 1 revolver and 1 normal undulator are already accounted for in proposed upgrades.

Note that it is likely that other optimized undulators are necessary to manage heat loads on older beamlines. These are not accounted for in this tally.

Several sectors could be canted but have not yet put forward a scientific case to do so. These undulators (and front ends) are likewise not included in our tally.

	Number not included elsewhere	Cost	Uncertainty Each
Revolver	1	2.5M	0.5M
Normal	2	3.0M	0.6M



APS Life Sciences Summary of Unfunded Upgrades (as of March 2013)

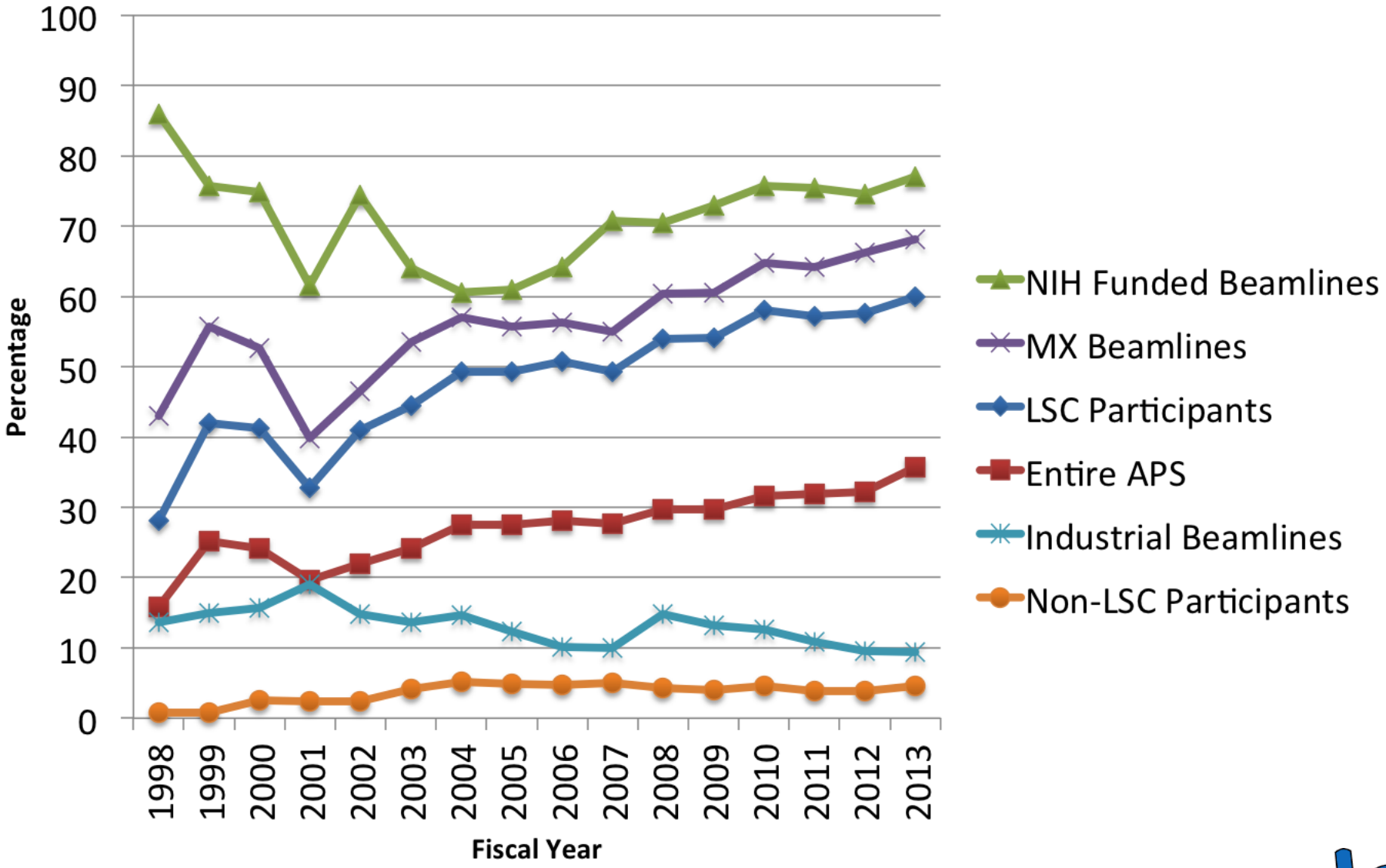
Category	Total ($M)	Uncertainty ($M)
Detectors	20.0	3.0
Undulators	5.5	0.8
BPMs	3.0	0.8
Facilities	34.7	4.6
Total	**63.2**	**5.6**

NOTES:

- Costs and uncertainties are very rough estimates
- All sections need a thorough cost audit before proceeding
- Uncertainty seems too optimistic for this stage
- On going annual costs: $2.0M +/- 0.3M

Missing from current estimates:

- APCF User Facility and other potential upgrades that are still early in the planning process
- Design stage funding (Projects differ wildly in their needs)
- Overhead costs and full effort


Percentage of APS Experiments with NIH Funding
Percentage
100
90
80
70
60
50
40
30
20
10
0
1998
1999
2000
2001
2002
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
Fiscal Year
NIH Funded Beamlines
MX Beamlines
LSC Participants
Entire APS
Industrial Beamlines
Non-LSC Participants

NIH Portion of APS Usage Comments

1. The portion of experiments run at the APS that rely on NIH funding is high and rising: more than 1/3 of all experiments conducted so far in fiscal 2013 have come with NIH funding.

2. The number of experiments is not the same as actual usage.

3. The graph data were derived from a table of experiments per sector per fiscal year, so the few programs that share a sector (NE CAT use of 8 BM, for example) are not portrayed accurately. However, it is unlikely that a precise accounting would change the conclusions significantly.

4. The NIH seems to be a likely funding agency for the non-industrial LSC beamlines.

Management

At this point the management of the new facilities would depend strongly on the funding mechanism. Not all host beamlines currently have the staffing needed to run and maintain more beamlines so, in addition to needed a fair bit of capital to construct these new capabilities, we'll need to find funding to operate and maintain them.

Possibilities:

1) A funding agency desires that the APS build and maintain some fraction of the LS-U beamlines.

2) Each CAT finds find their own funding source(s) for capital and maintenance.

3) Mixture of the above scenarios.

APCF Opportunity



The new Advanced Protein Crystallization Facility currently under construction has extra floor space available. There is an opportunity to start a new user facility in that space to provide protein crystallization support for the APS user community. The scope and scale of operations are still under discussions as is the extent that such a facility might be integrated into the planned APCF operations.

A minimal facility providing crystallization support for at least a portion of the user community would probably run something like $1.5M +/- $0.2M and a similar amount annually to run.

Thinking bigger and building a world class user facility to cover needs not met by the current APCF plans would cement the APS dominance of protein crystallography. This would run a bit more.